ZOO-506

Final Term paper 2022

Lion group for zoologist……

DEAR STUDENTS ,
IN 3ND SMSTER JOIN ADMIRE GROUP OF
ZOOLOGISTS FOR BETTER LEARNING.

GROUP PROPERTIES..

- SEPARATE GROUP FOR EACH SUBJE
- VIDEO LECTURES
- IDEAL ASSIGNMENTS
- IDEAL GDB

FINAL TERM AND MIDTERM PREPARATIO

FOR JOINING CONTACT WITH US
SIR ARSLAN = 0304–4142714

Sir A

FINAL TERM PAST PAPER

1. Damage cause by termites.

They enter through cut ends of setts & feed on the soft tissue.
The tunnel is filled with the soil.
The termites attack setts, shoots, canes and also stubbles Entire shoot dries up and can be pulled out easily




TERMITE

- **Workers** attack the newly planted setts and affect germination.



SYMPTOMS OF DAMAGE

- They enter through cut ends of setts & feed on the soft tissue.
- The tunnel is filled with the soil.
- The termites attack setts, shoots, canes and also stubbles
- Entire shoot dries up and can be pulled out easily



2. Intelleigent cultivation.

- Intelligent cultivation and use of fertilizers is well as pruning to remove injured or diseased parts will aid in insect control

3. Enlist the plants who do pollination

Some plants, as the Smyrna fig, clovers, melons, tomatoes, beans, peas and most of the common fruits, require the visits of insects before the seeds or fruits can form

OR

1- Okara
2- Potato
3- Onion
4- Mustard
5- Beet
6- Mango
7- Alfalfa
8- Apricot
9- Sweet cherry
10- Almond

4. Damage cause by red-cotton bug

Damage symptoms: The **damage** is caused by both nymphs and adults by sucking the sap of the plant as well as that of bolls and stain the lint. They are also, therefore, termed as **cotton** boll stainers. The **bugs** are gregarious in habit. Attacked seeds loose viability.

Sir Arslan



5. **What is hopper burn?**

Hopperburn or yellowing, browning and drying of plant. Affected plant dries up and gives a scorched appearance called “**hopper burn**”. ...



6. What is difference between cultivation and destruction of vegetable or clean culture.

Intelligent cultivation and use of fertilizers is well as pruning to remove injured or diseased parts will aid in insect control

3. 2- The **destruction of vegetable trash and weeds**, frequently called **clean culture**, is also an important factor, especially in the control of hibernating insects

7. What are attractants

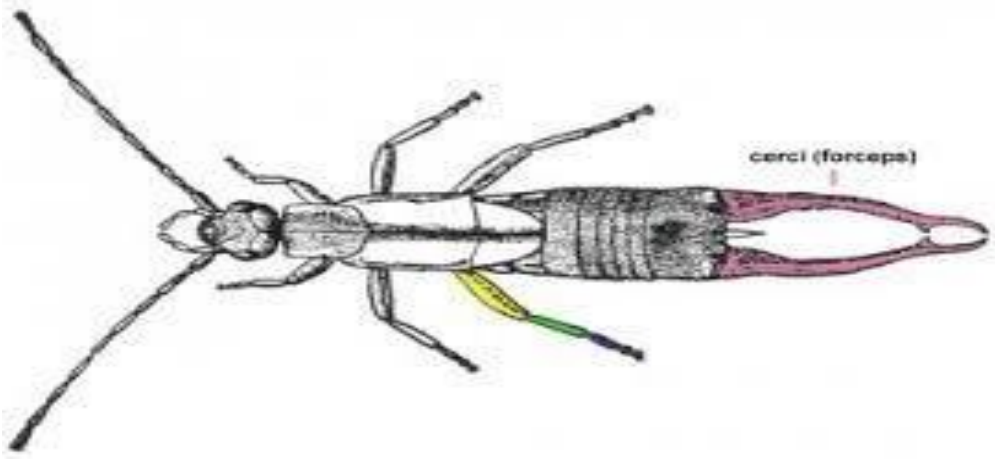
A chemical agent, usually associated with an **insect's** sexual drive, which may be used to attract pests to poisoned bait or for **insect** surveys





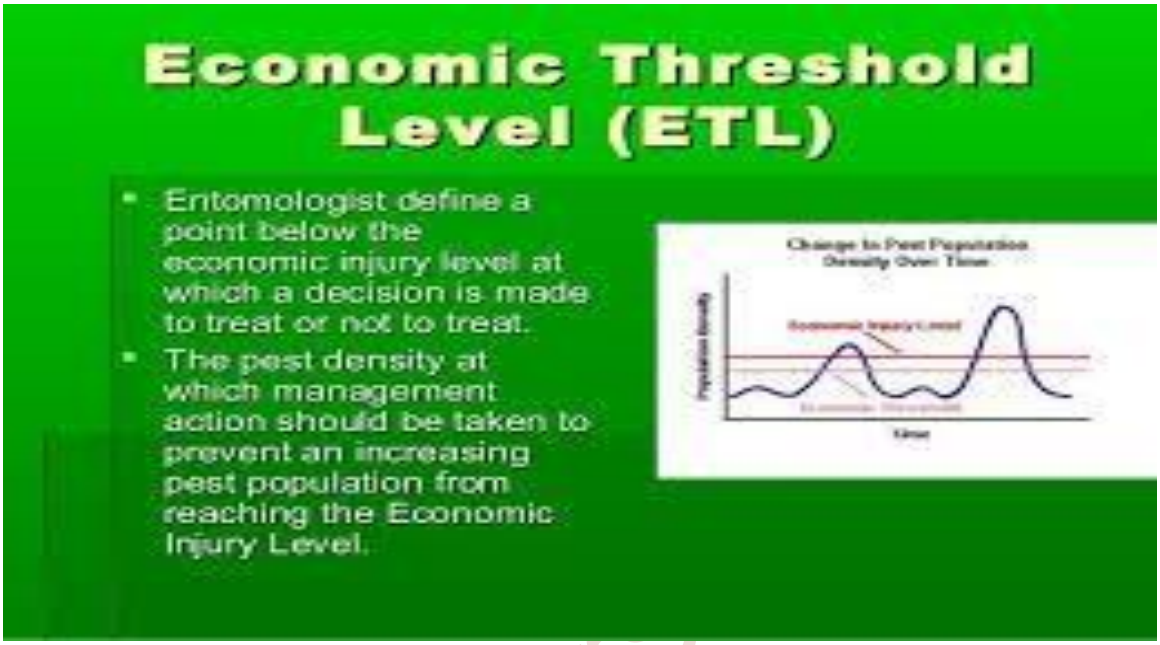
8. What is cerci?

A pair of many-segmented, antenna-like structures, sometimes short, sometimes long, may occur at the end of the abdomen and these are called cerci. They probably serve as organs of touch and possibly organ of smell in some cases



9. What is meant by threshold?

Economic threshold. The pest density at which management action should be taken to prevent an increasing pest population from reaching the economic injury



10. Field scouting .

Pest Monitoring, sometimes call **field scouting**, is an activity that can be accomplished by anyone with a little advanced training

OR

Field scouting is the regular examination of **fields** in a prescribed fashion to measure pest levels. To properly **scout** for pests, you must know where they live, what they look like, and how to find and count them. This

information is provided in the sections on specific pests.



11. **Define vactor.**

An agent that transports a microorganism from one host to another. Insects that transmit plant disease are called vectors.

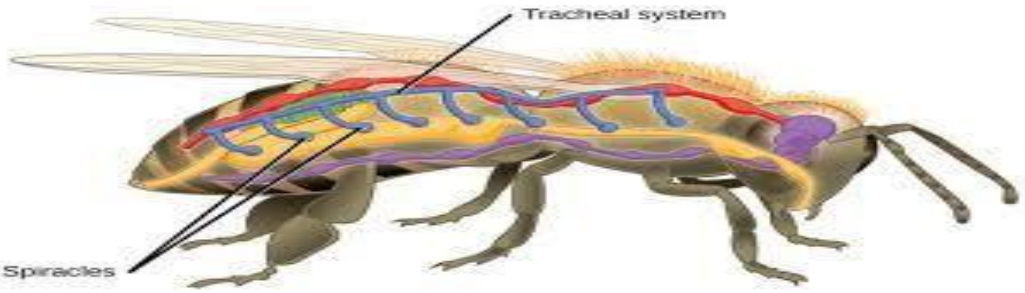
12. Tracheal gills

In **insects**, **gills** are usually outgrowths of the **tracheal** system. They are covered by a thin layer of cuticle that is permeable to both oxygen and carbon dioxide. ...

OR

Insects which in their early stage of live in water cannot, of course, breathe air into their bodies through spiracles during that period of their live There are enclosed in such cases and the animal obtains and usually through special structures called tracheal gills

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13. Family and order of army worm

Family: Noctuidae

Order: Lepidoptera.

It is a major and important chewing insect pest of cotton crop.



14. What is scientific classification

Scientific Classification is a system used to classify all living things through a breakdown starting with the largest grouping called a Kingdom.

2- And continuing down to the smallest grouping called species

15. Damage of ric-ear

(5) Symptoms of Damage

The pest is essentially diurnal with highest activities in the morning and in the evening, seeking shelter during hotter parts of day

Many generation are completed in a year

Sucking of the grain sap causes empty or partially filled and chaffy grains and enhances subsequent fungal and bacterial infection

OR

Rice bugs **damage rice** by sucking out the contents of developing grains from pre-flowering spikelets to soft dough stage, therefore causing unfilled or empty grains and discoloration.



16. Enlist isect-born diseases (5)

A few of the more important insect-borne disease in organisms are

- Malaria
- Bubonic plague
- Typhoid
- Typhus
- Yellow fevers.

17. Physical methods names (3)

Physical Factor Controlling Insects

1- Heat

2-Cold

3-Moisture

4-Light

5- Electricity

Note : topic 90 must read

18. : Enlist ways of integrated pest 5)

Utilizes all suitable pest management Ways

Pesticide Control

Cultural Control

Mechanical Control

Sanitary Control

Natural Control

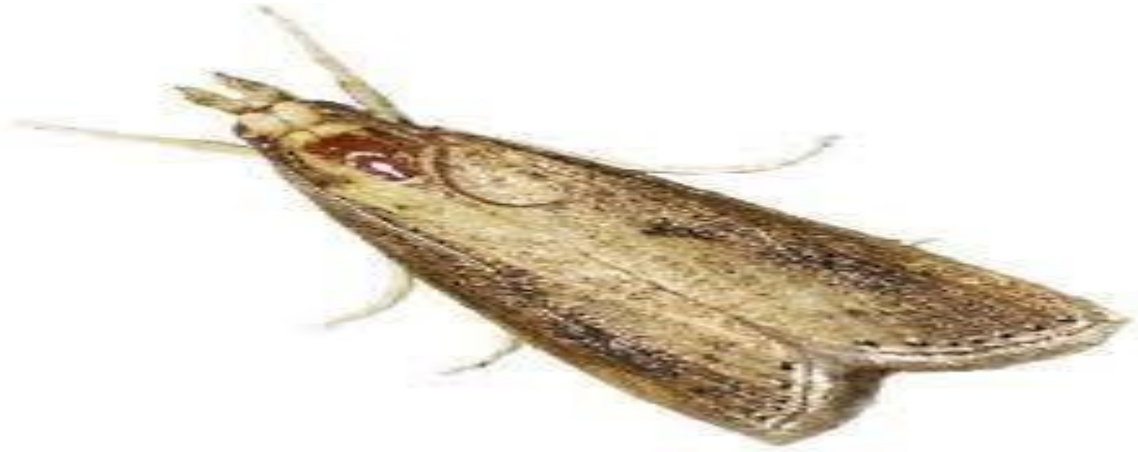
Biological Control υ Host Plant Resistance

19. **What is pest management?**

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Integrated **pest management** (IPM) combines the use of biological, cultural and chemical practices to **control** insect **pests** in agricultural production.It seeks to use natural predators or parasites to **control pests**, using selective pesticides for backup only
when **pests** are unable to be controlled by natural means

20. Scientific name of early shoot borre Chilo infuscatellus



21. Chemical control

The chemical control of insects includes control not only with **insecticides** in the true sense as discussed in the following chapters

But also with various materials used to **attract and repel insects** without actually killing them

OR

Chemical control is based on substances that are toxic (poisonous) to the **pests** involved. When **chemical** pesticides are applied to protect plants from **pests**, diseases or overgrowth by weeds, we speak of plant protection products ..... Then there are herbicides which are used against weeds.

22. What is physical control?

The extremes of both **temperature and humidity**, either high or low, are not only effective barriers in the natural control of insects

But they are also useful in artificial control

Light and electricity are other factors that come under this heading

OR

Physical control refers to mechanical or hand **controls** where the pest is actually attacked and destroyed. **Physical controls** are used mostly in weed **control**. Tillage, fire, removal by hand, grazing and mowing are all used to destroy weeds and prevent reproduction.

23. Biological control of white grub

Collection and destruction of adult beetles from host trees.

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Picking of grubs manually from field

Repeated ploughing & exposing various stages of grub to their natural enemies.

The use of light trap.



24. What is Host Plant Resistance?

Manipulating the crop to withstand or tolerate pests by

1- Natural breeding method

2- Genetically modified plants

Examples: Glandular-haired Alfalfa, Bt Corn

OR

Host plant resistance refers to "the heritable qualities of a cultivar to counteract the activities of insects so as to cause minimum per cent reduction in yield as compared to qualities of a cultivar to counteract the activities of insects so as to cause minimum per cent reduction in yield as compared to other .

25. . Two examples of arachnida. 2 marks

The Arachnida, including the scorpions, pseudo- scorpions, spiders, mites, ticks, etc.

26. What is legislative control? (2 marks

Various laws and regulations both Federal and state have been enacted to prevent the introduction of foreign pests:

a. To prevent their spread within the country

b. To enforce control and extermination

c. To ensure that chemicals used for controlling insects are neither adulterated nor misbranded

27. .what are repellents? Explain some repellents Repellents are those materials used to keep insects away from crops, animals and man

2. Various **coal-tar** and **pine** products are rather general insect repellents.

3. **Naphthalene** is a clothes moth repellent and oil of citronella is a common ingredient of preparations used to keep mosquitoes away from people

28. .write commercial products produce by insects..

A third group of insects is responsible for a number of commercial products

1-Silk

2-Honey

3-Beeswax

4-Shellac

29. Write the uses of pesticides.

Pesticides are used to control various pests and disease carriers, such as mosquitoes, ticks, rats and mice. Pesticides are used in **agriculture** to control weeds, insect infestation and diseases. There are many different types of pesticides; each is meant to be effective against specific pests

30. Mode of damage of fruit borer?



Bolls showing regular, circular bore holes

Larvae seen feeding on the boll by thrusting their heads alone inside and leaving the rest of the body outside

Presence of granular faecal pellets outside the bore hole.

A single larva can damage 30-40 bolls

OR

Sir Arsla

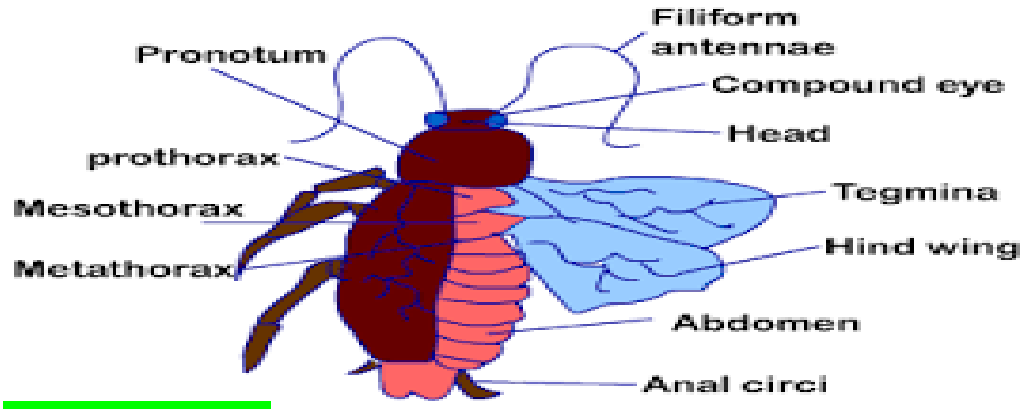
31. Write the name of thorax seqments.

The thorax has its three segments usually quite clearly marked

1- Prothorax

2- Mesothor

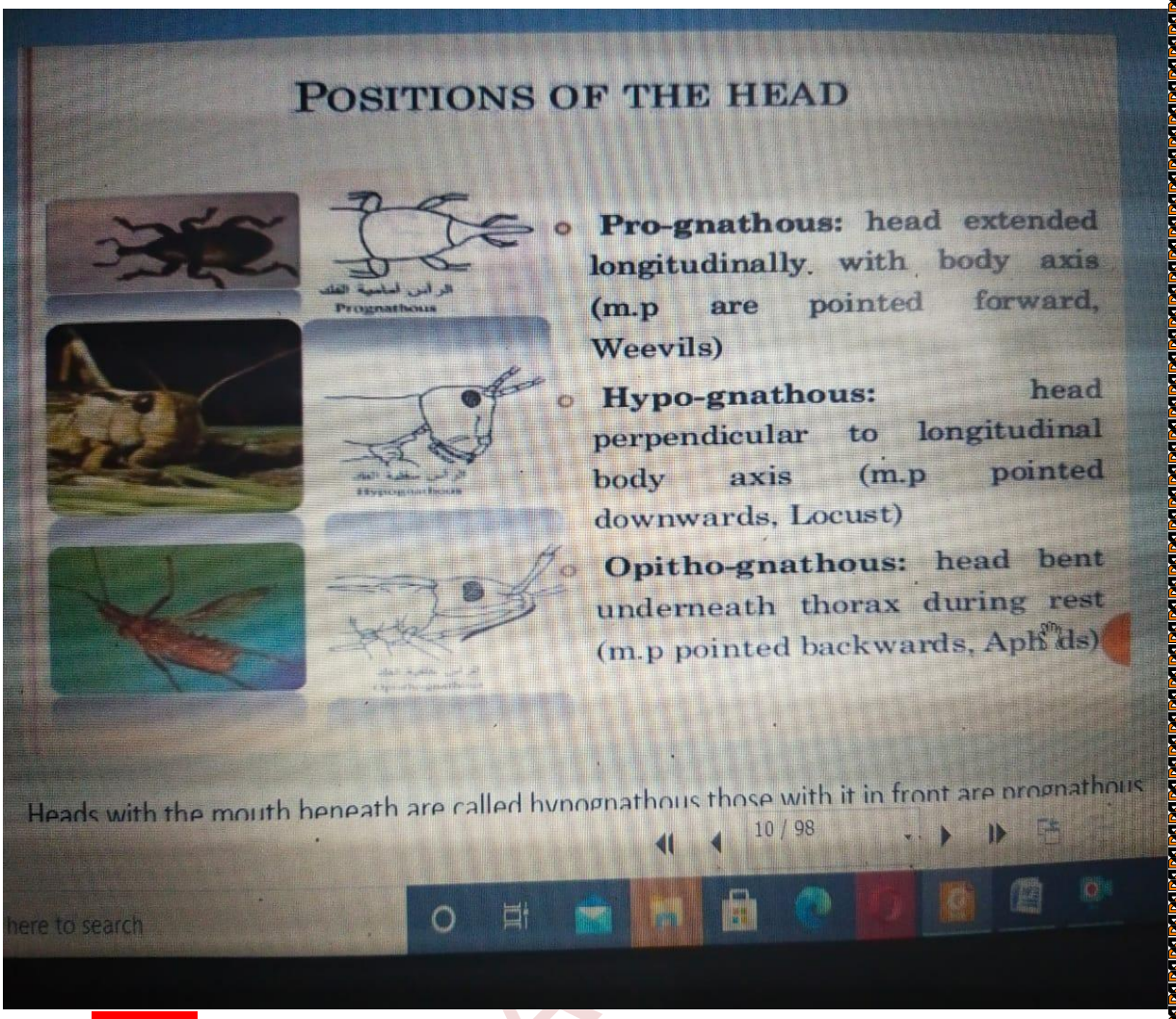
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Metathorax

32. position of head

Sir Arslan



33. Cold

Note low temperature pr question a rha hy but question ka nipta chla kis type ka a rha hy so nechy wala points just read kr len

Although **low temperatures** are not so effective as high ones in killing insects

2. Many insect pests of stored products such as **furniture, rugs, clothing and seeds** become inactive at **40 to 45°F.**

3. Under these conditions infested materials suffer no further damage or increase

infestation

4. And materials free from insects when put into cold storage will remain so.

5. Those insects which hibernate outdoors withstand temperatures far below zero

6. Most insect pests of stored products, however, are unable to become dormant in the true

sense and are killed in a relatively short time even at temperatures considerably above
zero

7. The use of low temperatures in the artificial control of insects, there fore, is of two types

note : is topic ko just read kren

34. Vector of

In the case of **malaria**, the **vector** is the anopheline mosquito and the disease-causing organism is the **malaria** parasite.

35. What is shell gland?

A gland-producing material, which forms the egg shell and is known as the shell gland, also opens into .this portion

36. Example of hemiptera

Examples: Assassin bugs, Kissing bugs, Leaf- footed bug, Lygus bug, Stink bug, Minute piratenbug,Big-eyed bug, Damsel bug

37. Phthirapter

Formally Anoplura and Mallophaga

Includes: Lice, Chewing Louse, Sucking Louse **Mouthparts**: Chewing and Piercing-Sucking **Metamorphosis:** Complete Metamorphosis

Order : Psocodea

Note: order or example examiner poch rha hy.



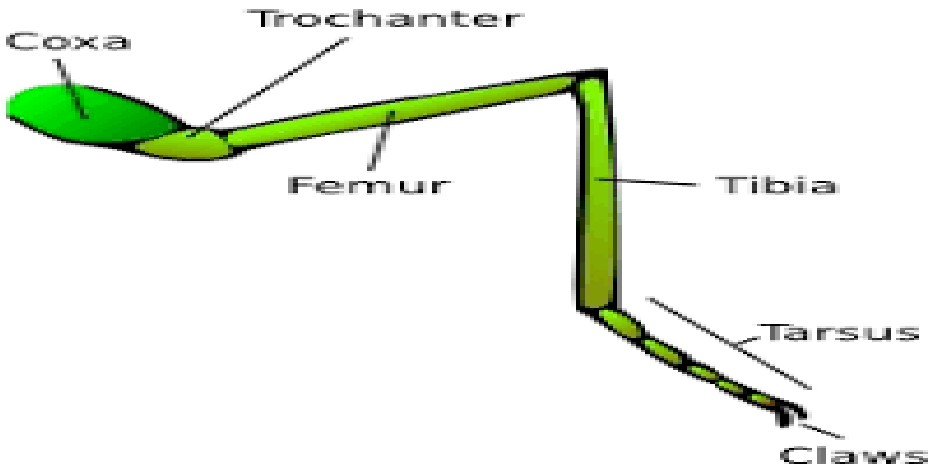
38. Transmission

Transmission is the passing of a pathogen causing communicable disease from an infected host individual or group to a particular individual or group .Regardless of whether the other individual was previously infected

39. Coxa

The coxa is the segment that articulates with the body, frequently partly lying in a more or less cupshaped hollow of the latter

It may be short or long and is generally freely movable on the body and powerful



40. Cause of allergy

Allergy (hay fever, asthma, etc.) are frequently caused by exposure of susceptible persons to dried insect scales and hairs

41. Classes of

1. **Class #1- Hexapoda:** Insects, six, jointed appendages
2. **Class #2- Chilopoda**- Centipedes, one pair of legs per body segment, multiple body segments
3. **Class #3- Diplapoda**- Millipedes, two pairs of legs per body segment, multiple body segments
4. **Class #4- Crustacea**- Crayfish, five pairs of jointed appendages
5. **Class #5- Arachnida**- Spiders, four pairs of jointed appendages

42. Diseas

. Any condition that impairs the normal functioning

of an organism or body. Or

Disorder of normal functioning of an organism

43. Epidemiology

The study of disease in populations of an organism.

4. Where live grasshopper of rice?

Various species of grasshopper are widely distributed in Nepal They are polyphagous and feed on leaves of rice, maize, millets, sugarcane, grasses etc

What things keep in mind during field scouting.5 marks

3. Realistically, a field scouting must know:
4. How to properly identify pests and their damage
5. How to get an accurate assessment of pest populations
6. A field scouts objective is to provide an accurate and unbiased objective ofpest populations
7. What tools and procedures do you use for alfalfa weevil for potato leafhopper, etc.
8. Both of these examples are pests on alfalfa, however, the sampling procedure is different.

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Order, family and symptoms of tobacco cutworm.5

Armyworm /Tobacco Cutworm: *Spodoptera litura*

- **Family:** Noctuidae **Order:** Lepidoptera.

- It is a major and important chewing insect pest of cotton crop.

Symptoms of damage

- Scrapping the epidermal layer, leaving the skeleton of veins of leaf
- During severe attack, only the stem and side shoots will be standing in the field without any leaf or bolls
- Larvae feed the leaves by making small holes.

Why foreign insects brought in the country.3

1. Various laws and regulations both Federal and state have been enacted to preventthe introduction of foreign pests:
    a. **To prevent their spread within the country**
    b. **To enforce control and extermination**
    c. **To ensure that chemicals used for controlling insects are neither adulteratednor misbranded**
2. Which in many instances have proved valuable in reducing the numbers of foreign pests
3. Probably the first introduction attempted from aforeign country was the now classic case of a **ladybeetle brought from Australia** to attack the **cottony cushion scale**
4. From abroad **to combat those foreign insect pests** brought to this country accidentally but without their natural enemies.

6 What is biological control 2 marks

Biological control or biocontrol is a method of controlling pests such as insects, mites, weeds and plant diseases using other organisms. It relies on predation, parasitism, herbivory, or other natural mechanisms, but typically also involves an active human management role.

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7 Method of biological control.5

Biological control methods

Biological control involves the use of insects or pathogens (diseases) that affect the health of weeds (called invasive plants in the Biosecurity Act 2014). Usually, these biocontrol agents are from the same country of origin as the weed species.

Biosecurity Queensland undertakes biological control research in Queensland. Strict measures are in place to ensure that these agents do not negatively affect native plants and animals or horticultural and agricultural crops.

Biosecurity Queensland's biological control group is presently working on 10 weeds of concern to Queensland:

- bellyache bush
- cat's claw creeper
- lantana
- madeira vine
- mother-of-millions
- parthenium weed
- prickly acacia
- siam weed
- cabomba
- opuntioid cacti.

A number of biological controls have been released at strategic infestation sites, and are spreading to other infestations of their host plant.

Biological control agents can reduce the vigour, size and competitiveness of weed infestations; however, they rarely get rid of them altogether. Biological control works best in conjunction with other control methods

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8 Appearance of adult rice-ear.2

Rice ear- head bug

- The adult of *Leptocorisa oratorius is green, light brown or* mixed yellow in color

The adult are slender and about 20mm long.

9 Define arthropods? 2

An “arthropod” is an invertebrate animal that has an exoskeleton, a segmented body, and jointed appendages

any of a phylum (Arthropoda) of invertebrate animals (such as insects, arachnids, and crustaceans) that have a segmented body and jointed appendages, a usually chitinous exoskeleton molted at intervals, and a dorsal anterior brain connected to a ventral chain of ganglia

10 Damage by termites?3

Termite

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- *SYMPTOMS OF DAMAGE*
- They enter through cut ends of setts & feed on the soft tissue.
- The tunnel is filled with the soil.
- The termites attack setts, shoots, canes and also stubbles Entire shoot dries up and can be pulled out easily

11 What are attractant? 3

Attractants

1. Sometimes called **attrahents** are materials used to **lure insects** to traps and poisoned baits
2. The use of geraniol in **Japanese beetle traps**, **sugar sirups** for **ant baits**, and **fermenting sirup solutions** for trapping various moths are good examples.

12 Advantage of weeds destruction in cultural control? 3

1. **2-** The **destruction of vegetable trash and weeds**, frequently called **clean culture**, is also an important factor, especially in the control of hibernating insects
2. Weeds not only interfere with successful crop growth by **crowding**, but they may reduce the vigor of the crop by **competing for plant food** in the soil
3. In winter they provide **hiding places for hibernating pests**
4. A pest appearing in the spring before its crop food plant is available may find certain weeds palatable until the crop has appeared
5. Other pests move to late summer and fall weeds after their particular crops havebeen harvested

It largely involves manipulating farming practices to suppress **weed** growth and production, while promoting the development of the desired plant. The principles and techniques used to prevent **weed** spread are relevant to **cultural control**methods to limit the spread of **weeds** between different land areas.

13 Time of planting of sugarcane for cultural control? 5

IPM for Sugarcane borers(Cultural Control)

- **Time of planting**
- Early planting during December –January to escape the shoot borer incidence.
- Detrashing
- Internode borer and Stalk borer
- Manuring
- Avoid excess use nitrogenus fertilizers.
- Stalk borer and internode borer.

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- Trash mulching and earthing up
- Removal and destruction of infested cane
- Early shoot borer
- Collection and destruction of eggs
- Internode borer and Top borer

14 Explain Biological control?5

1. **Parasitic and predaceous** insects as well as diseases of insects have already been discussed
2. As vital factors in the **natural control** of injurious insects
3. It is possible, however, for man to encourage such beneficial forms by introducing them into new areas
4. By growing them artificially for liberation whenever they become scarce in nature
5. Until recent years most of this work dealt with the introduction of parasites
6. From abroad **to combat those foreign insect pests** brought to this country accidentally but without their natural enemies
7. Certain **native parasites and predators** are now being reared in large numbersfor liberation in areas badly infested with their host species
8. Probably the first introduction attempted from a foreign country was the now classiccase of a **ladybeetle brought from Australia** to attack the **cottony cushion scale**
9. Which had already reached California and was endangering the existence of the citrus industry
10. Following this successful attempt many other parasites and predators have been imported
11. Which in many instances have proved valuable in reducing the numbers of foreign pests
12. Parasites should not be expected to kill off all their hosts
13. They will merely reduce the population to greater or less extent
14. Other control measures must also be used if the insect is to be exterminated
15. The discovery and introduction of parasites are expensive
16. However, since a successful parasite perpetuates itself, the initial high costof establishment may be spread over a period of many years
17. Many dangers are involved in bringing parasites into a new country
18. These insects may attack not only the pest that is to be controlled
19. But also other parasites of the pest
20. In such cases the introduction may be of little value

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21. There is always the chance also that their own secondary parasites may be imported along with the primary ones

22. Only thoroughly trained entomologists

23. Therefore, should be permitted to bring parasites and predators to this country if we are to avoid the introduction of undesirable insects along with the desirable ones

24. The liberation of insect diseases is not so generally successful as variations in natural conditions limit the growth of the disease

25. Many of the disease organisms are widely distributed and appear spontaneously

26. When suitable conditions prevail, even without any encouragement

27. The most systematic use of diseases in artificial insect control in the United States appears to be the annual distribution by the state of Florida of **fungi** that live on the **whiteflies that attack citrus**

15 Economic importance of insects? 5

A. Of the many thousands of insect species known to man some affect him and hisproperty only slightly; some are beneficial; others are injurious in various degrees

B. Only rough estimates can be made of the monetary losses and gains by insectactivities and even these are made with difficulty

C. Although it is the general practice to express losses in terms of dollar valuation

D. It probably would be safer to give them in bushels, tons, etc., because the relative values of different crops and products change from year to year

E. No crop of either vegetable or animal origin appears to be entirely free from insect injury

F. Field, truck and fruit of crops, both growing and in storage, household goods and food products, forests and the wood products derived from them, domestic animals and animal products, all are more or less liable to insect attack

G. Because there never has been a season free from the ravages of insects which mightserve as a standard for comparison, much of this loss is not appreciated

H. If there could once be such a year entirely insect free, the difference would be apparent at once

16 How shot and top and internode borer can be biologically controlled?10

Biological Control

 For Early shoot borer :

 Trichogramma chilonis **For Top borer :**

Trichogramma chilonis or T. japonicum, Isotima javensis **For Internode borer :**

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- *Trichogramma chilonis*
- Cultivation of resistance variety
- Irrigation at closer intervals for managing
- Early shoot borer
- Practice deep harvesting to destroy stubbles
- Removal of water shoots to destroy
- Top shoot borer

17 Sanitary control 3

What is Sanitary Control?

- **Methods to avoid introducing a pest into a field by**
- 1- Cleaning field equipment
- 2- Planting certified seed
- 3- Quarantines

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18 Economically importance of insects 2

- The benefits to man from their activities are frequently overlooked in summing up the economic importance of insects
- Although in many cases it is impossible to say how much good is derived from insects 1-Attack on injurious species
- Reliable statistics are easily obtained relative to the production of useful insect
- products Those insects which attack injurious species probably form the most important group

They are divided into two sections:

- 1-The parasites that live in or on the different stages of other insects, then called hosts
- 2-The predators that capture and devour other insects
- Parasites usually, though not always, live in or on a single host insect during their entire development
- Whereas a single predator may derive its nourishment from many victim insects.
- Parasites are of many species, frequently so specialized in their habits that one kind of parasite attacks only one or a few host species
- In some cases the parasites work only in insect eggs and thus prevent any feeding loss by the host insect
- Other parasites attack early stages of the pests, in some instances killing them soon

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- In others not until the pests have completed their feeding in spite of being parasitized.
- In the latter the benefit is less, but at least there are fewer adults to produce the next generation
- All stages are known to be attacked in different species by one parasite or another

Pollination

- Some plants, as the Smyrna fig, clovers, melons, tomatoes, beans, peas and most of the common fruits, require the visits of insects before the seeds or fruits can form
- Another highly important group of insects includes those which pollinate flowers during their visits for nectar and pollen, thus enabling the plants to produce seeds
- The value of these crops which require the services of insects in the United States amounts to over two billion dollars

3- Commercial Products

- A third group of insects is responsible for a number of commercial products
- **1-Silk**
- **2-Honey**
- **3-Beeswax**
- **4-Shellac**

4- Diet of certain birds, mammals

- Insects provide the entire diet of certain birds, mammals, etc., and an important part of the diet for many others

5- Serve as Food

- Even man in some parts of the world consumes large quantities of grasshoppers, crickets, ants and other insects
- The great value of some insects and the great losses, sometimes enormous, caused by others make it obviously important to encourage the beneficial forms while striving to prevent or at least to reduce the losses caused by the injurious ones

19 Advantages of weeds destruction 3

6. **4- The destruction of weeds and brush** along fences is of advantage in the reduction of insects
7. But it results in the elimination of nesting and hiding places for insectivorous and game birds

20 Biological control 5

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54. The most systematic use of diseases in artificial insect control in the United States appears to be the annual distribution by the state of Florida of **fungi** that live on the **whiteflies that attack citrus**

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21 Order of termites 2

Order: Blattodea

Isoptera

- **Includes:** Dry-wood termite, Subterranean termite
- **Mouthparts:** Chewing Mouthparts **Metamorphosis:** Incomplete Metamorphosis
- Live for 15 years.
- Lay 1 egg every 15 seconds
- Eat wood. Can destroy a house in 2-3 years
- Found in every US state except Alaska.
- Recycle wood in the soil

22 Pests of rice 10

PESTS OF RICE

Rice (*Oryza sativa* L.) is the most important staple food crop of the world

- More than 60 % of worlds population depend on it for food.

Major insect pests

- **1-Brown Plant Hopper**, *Nilaparvata lugens* (Homoptera: Delphacidae)
- **2-Yellow Rice Stem Borer**, *Scirpophaga incertulas* (Lepidoptera: Pyralidae)
- **3-Rice Ear-head Bug**, *Leptocorisa oratorius* (Hemiptera: Alydidae)
- **4-Rice Hispa,** *Dicladispa armigera (Coleoptera:* Chrysomelidae)
- **5-Rice Grasshopper**, *Hieroglyphus banian* (Orthoptera: Acrididae)

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23 Family/ Order Symptoms of damage of Tobacco Cutworm 5 marks

Armyworm /Tobacco Cutworm: *Spodoptera litura*

Family: Noctuidae **Order:** Lepidoptera.

- It is a major and important chewing insect pest of cotton crop.

Symptoms of damage

- Scrapping the epidermal layer, leaving the skeleton of veins of leaf
- During severe attack, only the stem and side shoots will be standing in the field without any leaf or bolls
- Larvae feed the leaves by making small holes.

24 Low and high temperature in artificial control

- Heat

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2- Most insects cease reproduction and soon die at temperatures rom **100 to 1l0°F**

3- The temperature of their surroundings should be raised to **120° or higher** to kill them with heat artificially **within a few hours**

4- The length of period for effective treatment is more dependent upon the

   a. time to heat the grain

   b. clothing

   c. other material in which the insects are located

5- Than upon the heat resistance of the pests themselves

6- Because no insect can survive more than an hour if actually exposed to

atemperature of 120°F 7- Bales and bags are difficult to heat through

cold

1. Although **low temperatures** are not so effective as high ones in killing insects
2. Many insect pests of stored products such as **furniture, rugs, clothing and seeds** become inactive at **40 to 45°F**.
3. Under these conditions infested materials suffer no further damage or increase in infestation
4. And materials free from insects when put into cold storage will remain so.
5. Those insects which hibernate outdoors withstand temperatures far below zero
6. Most insect pests of stored products, however, are unable to become dormant in thetrue sense and are killed in a relatively short time even at temperatures considerably above zero
7. The use of low temperatures in the artificial control of insects, there fore, is of two types
8. The more important is cold storage, as for furs, at temperatures just low enough toprotect the stored materials from injury
9. The other use of cold actually to kill insects is feasible only on a large scale in zeroweather in the north
10. Here as in the case of the heat-treatment much more time is required to chill grain, flour, clothing and other products than is usually realized

25 Fuzz purposes

Fuzz: It is used in production of mattresses, surgical cotton, photographic film and paper.

26 Scientific name of early shoot borer

- **For Early shoot borer :**
- *Trichogramma chilonis*

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What is Biological Control?

Manipulation of biological organism to control pests by:

- 1- Release of predators/parasites/disease of an insect or weed Can be time consuming, expensive and difficult **What is Host Plant Resistance?**
- Manipulating the crop to withstand or tolerate pests by
- 1- Natural breeding method
- 2- Genetically modified plants
- **Examples**: Glandular-haired Alfalfa, Bt Corn,

. where Rice Grass hopper live 2

- Various species of grasshopper are widely distributed in Nepal
- They are polyphagous and feed on leaves of rice, maize, millets, sugarcane, grasses etc

Which machinery is used to control pest 2

- Uses machinery and/or other tools to control pests
- 1- Tillage: Tillage is the agricultural preparation of soil by mechanical agitation of various types, such as digging, stirring, and overturning

Symptoms of damage of **Fruit borer** 3

- Bolls showing regular, circular bore holes
- Larvae seen feeding on the boll by thrusting their heads alone inside and leaving the rest of the body outside
- Presence of granular faecal pellets outside the bore hole.
- A single larva can damage 30-40 bolls.

Why Agronomic practices that are designed to: 3

- 1-Optimize growing conditions for the crop
- 2-Anything that increases a crop's competitive edge will result in increased tolerance to pests often resulting in reduced

pesticide use 3- Create unfavorable conditions for the pest

1- Describe the places where heat treatment is used to kill the insects 3

2- Most insects cease reproduction and soon die at temperatures rom **100 to 1l0°F**

3- The temperature of their surroundings should be raised to **120° or higher** to kill them with heat artificially **within a few hours**

4- The length of period for effective treatment is more dependent upon the

a. time to heat the grain

b. clothing

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c. other material in which the insects are located

5- Than upon the heat resistance of the peststhemselves

6- Because no insect can survive more than an hour if actually exposed to atemperature of 120°F 7- Bales and bags are difficult to heat through.

MID TERM MCQS

1. Scorpions, spiders, mites and ticks are the members of the group of arthrapods._____Arachnids

2. Crustaceans breathe through_____Gills and body surface

3. Which one of them is NOT the member of phylum Mollusc_____Sea urchin

4. In insects system is concemedwiththe nervesupply ofthealimentary canal_____-__Sympathetic nerve

5. Members of theundergogradualandusuallyincomplete metamorphosis._____Holometabola

6. is a minute opening or group of openings through the shell___-_____Micrapyl

7. Testes arepresentinthefront…….partoftheabdomen._____upper

8. The young in the Hemimetabole are called_____Naiads

9. Thefeedingperiads betweenthemalts(or ecdyses) arecalled_____instars

10. Presence of jointed legs is the characteristic of_____Arthropod

11. In insectsthere are……legs attached to the thorax_____6

12. Whichofthefollowingorganismhas Booklungas respiratory organ?_____Scorpion

13. Each ganglion is connected by one or two transverse bundles of nerve fibers which are called_____Commissures

14. The…… is the ventral portion of a segment of an arthropod._____Stemum

15. A sort of pad or cushion found between the pair of claws is called?_____Puvvillus

16. Insects have…… pair/pairs of wings._____two

17. The tracheae are lined by ……connected with that of the surface of the body._____Chitin

18. Insects having practically no metamorphosis are the member of the group……._____Ametabola

19. Nodus and…… are parts of insects wings_____pterostigma

27. Insects have …… types of eye._____Two

20. ……. have a flatter body while……are more rounded._____Centipedes, millipedes

21. Class Diplopoda include…….._____--Millipede

22. Larvaeofbeetles areusuallycalled_____grubs

23. Oviduct extends downward and back- ward around the side of the….._____Alimentary canal

24. The number of ocelli in insects is……._____1-3

25. When the body segments are fused with the head they form……._____Cephalothorex

26. Below the esophagus is a second ganglion, called the……_____Sub esophageal ganglion

27. Insects have …… types of eye._____Two

28. The stomach may develop side pouches called_____Gastric caeca

29. The tubes in testis unite to form 4 single tube, the……._____--------Vas deferens

30. Incentipedes, eachsegment bears …… pair/pairs of legs._____Only one

31. Thread or silk is spun backward and forward and around the body until it sometimes forms a complete outer covering called_____Cocoon

32. Insects have only……system of bloodvessels._____Incomplete

33. In …… the front leg on each side is modified to serve as a poison claw. Arachnids-_____chilopoda

34. Sow bug is the member of class……._____Gastric caecaGizzard

36. In insects heart is present in the …… part.Abdomen

37. The ……are the areas of membrane between veins _____cells

38. Headswiththemouthbeneatharecalled_____hypognathous costa

46. The…… is modified ovipositor, which is attached to poison glands._)_____Cerci

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47. In .....insect hatches from the egg practically in an adult condition.-_____Ametabola
48. Adultinsects…….haveto .....segments._____7 - 15
49. The .....are the areas of membrane between veins.chambers_____cells
50. In many cases, sexual organs not having become fully developed such insects are termed…._____Neuters
51. When the insect escapes from the cocoon skin it is now called a_____Pupa
52. In insects Cuticle is present outside the……_____Hypodermis
53. There are ..... segments in the abdomen of the adult insect._____3-11
54. Insome insects whochanged their habit to make holes for egg laying, the ovipasitorbeing a…__sting
55. Behind each mandible at the side of the mouth is theLabrum
56. The cuticle of the insects is largely made up of…Chitin
57. At the front end of the heart finds itself in a tube without chambers or valves, called_____Aorta
58. Whichofthefollowinginsecthasprognathoushead?_____Weevil
59. Smalldiplopods arefoundboringintopotatoesandsometimeswronglycalled)_____Wireworm
60. The…. extends from the mouth tothe anus atthe hinder end._____Alimentary canal
61. In insects Antenna are present in.....region_____Head
62. Crustaceans breathe through……..._____Gillsandbodysurface
63. The tubes in testis unite to fom a single tube, the_____Vas deferens
64. In some insects. eggs remain in the body of the parent for a time after hatching are_____Ovoviviparo
65. Scorpions, spiders ,Mites and ticks are the members of the group..... of arthropods.____Arachnid
66. Whichtype ofskeleton is found in arthropod?_____Exoskeleton
67. Oviductextendsdownwardand back-wardaroundthesideofthe…..._____Alimentary canal
68. In nervous system the cell bodies of neurons gathered together in clusters which are called)_____Ganglia
69. Thefirstabdominalsegmentiscalledthemediansegmentor_____Ropadeum
70. Belowthe esophagus is asecond ganglion.calledthe_____Sub esophagealganglion
71. The..... is the ventral portion of a segment of an arthropod._____Sternum
72. Near the base of the labium on its inner side there is a tongue like structure covered by_____Hypopharynx
73. In some insects, eggs remain in the body of the parent for a time after hatching are termed Viviparou_____Ovoviviparous
74. Insects have .....symmetry._____Bilatera
75. Insects have……. pair/pairs of antennae on their head._____One
76. In centipedes, each segment bears…… pair/pairs of legs._____------Only one
77. The ..... is usually small and may not be visible on all sides of the leg_____Trochanter
78. Which ofthe following is NOT the function of cerci?_____Used tostore eggs
79. The sclerites on bothsides of the segments present mainly in thoracic region arecalled_____Pleuron
80. The stomach may develop side pouches called_____Gastric caeca
81. Nodes and.....are parts of insects' wings._____pterostigma
82. Each ganglion is connected by longitudinal tract of nervefibers called_____Commissures
83. The nervous system of insects is located along the.....line of the body.Stream_____Middle
84. The ingrowth from the anus iscalled_____Hind intestine
85. In…… insect hatches from the egg practically in anadult condition._____Ametabola
86. Shellglandalsoopensinto_____Seminal receptacle
87. When the body segments are fused with the head they for_____cephalothorax
88. The walls of Malpighian tubes consist of asingle layer ofcellssurrounding a central channel which at one end open into the_____Hind intestine
89. In the female insect the eggs are produced in a pair of….. located in the upper frontpart of the abdomen_____Ovaries
90. The …..areslight grooves orscratches onthesurface forming the movable joints._____Sutures
91. The..... is a tube-like organ at the end of the abdomen and is used for the laying ofeggs._____Ovipositor
92. In adult insects ,number ofsegments in thorax region is_____ 3
93. Shedding of cubicula to let the organism grow is called_____Molting
94..... are very rarely present on the abdomen in adult insects but are often found in the_____Legs/
95. System of blood vessels coming to an end in..... Region._____Abdomen
96. When the insect escapes from the cocoon skin it is now called a_____Pupa
97. Members of the .....undergo gradual and usually incomplete metamorphosis._____Paurometabola
98. The group of insects that supply nourishment to the…. before its birth are termed_____Young
99. In insects,certain amount ofpoison can be eliminated from the body by the_____Malpighian tubes
100. WhichofthefollowinginsecthasOpithognathoushead?_____Aphids

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101. The .....forms the stomach of the adult insect._____Mid intestine
102. In female insects, ovaries are located in the upper front part of the_____bdomen
103. The sclerites on both sides of the segments present mainly in thoracic region arecalled_____Pleuron
104. In some flies the larva shrinks within its larval skin and transforms, this skin iscalled..... functioning like a ocoon._____Puparium
105. Total number of segments in the insects embryo is
106. Which ofthe following is the part of hind intestine?_____Rectum
107. In .....Heads, labrum is present above the mouth opening._____Prognathous
108..... is the Respiratory structure in Insects._____Book lungs
109. The compound eyes are also called .....Eyes._____Ocelli
110. Which of the following part of the mouth is concerned with the sense of smell?_____Maxillary palpus
111. Saltand..... aretransportedintothelumenofMalpighian tubule fromhemlymph._____Uric acid
112. A pair of jaws present behind the labrum is called_____Mandible
113. The hind intestine is divided into different parts which are..... Colon and_____ileum
114..... generally lie in the front of the thorax and open into the mouth_____Salivary gland
115. The tip of the wing is called_____Radiate
116. The hinder lip is called_____Labium
117. The mandibles and..... move sideways_____Maxillae
118. InMillipedes, eachsegment..... bears pair/pairs oflegs._____Two
119. In some flies the larva shrinks within its larval skin and transforms, this skin is called.....functioning like acocoon_____Puparium
120. The segments of the body of Arachnids are grouped into ..... sections._____Two
121. Larvaeof flies are usuallycalled_____Maggotss
122. Whichtype ofskeleton is found in arthropod?_____Exoskeleton
123. Which part of leg is quite long, moreslender thanthe femur and often providedwith downwardly projecting spines?_____Tibia
124. In nervous system the cell bodies of neurons gathered together in clusters which are called_____Ganglia
125. There are ..... segments in the abdomen of the adult insect._____3-11
126. In the egg the energy for development is provided by the....._____Yolk
127. Usually insects have..... pairs of wings._____2
128. Which of the following is NOT the stage of holometabolous life cycle?_____Nymph
129. In insects Cuticle is present outside the_____Hypodermis
130. In adult insects, number of segments in thorax region is_____3
131. Which of the following part ofthe thorax does not bear wings?
Prothorax
132. Shellglandalsoopensintr_____Seminal receptacle
133. Thetracheaefrequentlyenlargeandforms_____Air sacs
134. Holometabola are those insects which on hatching are totally ..... the adults and become adults by a somewhat different process+_____Unlike
135. Class Diplopoda include_____Millipede
136. WhichofthefollowinginsecthasOpithognathoushead?
_____hids
137. The third part of the thorax is called_____Metathorex
138. In the.....threads attach the hinder end of the body form a supporting loop
Butterflies
139. In adult insects, number of segments in thorax region is_____3
140. If the head is extended longitudinally to the body axis it is..._____Prognathous
141. The ingrowth from the mouth is usually called_____Fore intestine
142. The hinder lip is called_____Labium
143. A larva transforming in tunnels in wood forms a_____Partial cocoon
144. In embryo of insects number of segments in the head region _____6
145. The front lip is called_____Labrum
146. The .....are slight grooves or scratches on the surface forming the movable joint_____Suture
147. Unfolding ofthe imaginal buds leads tothe formation _____All
148..... are used for capturing other insects._____Forelegs
149. The ingrowth from the anus is .....called_____Hind intestine
150. The lobsters,crayfish,crabs,scorpions, spiders,mites, ticks,centipedes and millipedes are members of ..... group_____Insecta
151. Mammals belong to phylum_____Chordata
152. The sutures and the joint lines divide the cubicula into areas called plates or....._____Sclerites
153. Malpighian tubes function as a humn_____Stomach
154. Thejumping power of the grasshopper is duetothe great development_____Hind legs

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155. Insectswhichundergocompletemetamorphosis arecalled_____Hemimetabolous
156. The insect from hatching until maturity. is call_____Nymph
157. In .....Insect hatches from the egg practically in an adult condition._____Ametabola
158. Crop and Proventriculus are the parts of_____Mid intestine
159. In insects, the cells by their activities produce not only carbon dioxide gas but also_ waste material_____Nitrogenous
160. The openings by which the air enters the insects body are called_____Spiracles
161. ……have a flatter body while … .are more rounded._____Centipedes, millipedes
162. Tiny antenna- likestructure consisting of one tosix segments and attached tothe maxilla is called_____Maxillary palpus
163. The size of ocelli is equal to the one facet of eye_____Compound eye
164. The…… extends from themouthtotheanus atthehinder end._____Alimentary canal
165. The young in the Hemimetabola are called_____Naiads
166. A particle of blood containing nitrogenous waste material will wash over certain_____Malpighian tubes
167. Some cave-inhabiling insects have no_____Eyes
168. In the flies the hinder pair is modified and forms small structures called_____Halters
169. Ovipositor consists of …. Moveableparts._____3
170. After the completion of the construction of adult structures the …… skin splits and sets free theinsect._____pupa
171. Millipedes arefound onlandandbreathethrough_____Air tube
172. The young's in the Paurometabole are called_____Nymphs
173. Heads with the mouth beneath are called._____hypognathous
174. There are....types of metamorphosis._____3
175. Each spiracle opens into a short tube called_____Trachea
176. Insects have……. symmetry._____Bilateral
177. In insects heart is present in the ….pa_____Abdomen
178. In Arachnids number of legs attached to the cephalothorax is_____8
179. The third part of the thorax is called_____---{Metathorax
180. In female insects, ovaries are located in the upper front part of the_+_____bdomen
181. The largest group ofanimals is_____Insecta
182. In adult insects number ofsegments in thorax region is_____1
183. Theinsectbloodconsists ofcorpuscles whichresembles theofhumanblood_____None of these
184. Insects which lay eggs aretermed as-_____Oviparous
185. Thelobsters,Cray-fish,crabs,scorpions.Spiders,mites.Ticks.Centipedesand millipedes are members ofgroup._____Insecta
186. Whichof the following insects havedivision of labor?_____All of the above
187. Near the base of the labium on its inner side there is a tongue like structure covered by bristles or hairs called_____Hypopharynx
188. The young in the Hemimetabole are called_____Naiads
189. A few insects that supply nourishment to the young before its birth are termed_____Viviparous
190. An enlarged portion of the vas deferens used for the temporary storage of the_____Seminal vesicle
191. After the completion of the construction of adult structures …… the skin splits and sets free theinsect._____Pupa
192..... are very rarely present on the abdomen in adult insects but are often found in theearlierstages._____Legs
193. Sucking insects are the agents of .....transmission+_____Biological___Hinder
195. In some insects maturity of the..... organs is not complete until a little later_____Reproductive
196. Which one of them is NOT the member of phylum Mollusca?_____Sea urchin
198. In the .....Upon hatching a feeding period or instar comes first, followed by a molt and growth_____Holometabola

199. In the front leg on each side is modified to serve as a poison claw._____Chilopoda
200. Insects have .....pair/pairs of wings._____Two
201. A sort of pad or cushion found between the pair of claws is called+_____Pulvillus
202. A pair of jaws present behind the labrum is called_____Mandible
203. The tracheae are lined by connected with that of the surface of the body_____Chitin
204. Insects which in their early stage of live, in water breathe through special structures calle_____Tracheal gills
205. Which of the following is NOT the stage of hemi metabolous life cycle?_____Pupa
206. Which of the following part ofthe thorax does not bear wings?_____Prothorax
207. The largest ganglion of the body above the esophagus. produced by the fusion af several ganglia is called_____Brain

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208. Insectshaving practically nometamorphosis arethemember ofthegroup_____Ametabola
209. Behind each mandible at theside ofthe mouth is the_____Labrum
210: In the Hexapoda the segments of the body are grouped ……in distinct sections_____Three
211. Alarvatransforming intunnelsinwoodforms a……_____ Partial cocoon
213. There are numbers of membrane in each wings of the insect.
219..... Is located at the inner surface of labrum and forms the roof of the mouth.----------Epipharynx
220. In insectsthere are …….. legs attached to the thorax._____2
221..... are the most successful, widespread and prolific members of the animal_____Insects
222. Seminalreceptacle attachedtovagina is usedtostore_____Sperms
223. Insects have only system of blood vessels_____Incomplete
224. Which ofthe following part of the mouth is concerned with the sense ofsmell?_____Maxillary palpus
225. And the skeleton in arthropoda is made up of ……. which somewhatresembles horn in its nature._____chitin
226. Bees. Wasps, Ants are included in order of class hexapoda ._____Hymenoptera
227. The length of period for effectivetreatment is more dependent upon the time to heatthe grain clothing other material in whichthe insects are located_____all
228.Insects belong in the_____Arthropoda
229. Beetles and Weevils are included in order of classHexapoda._____Coleoptera
230. Centipedes arethemember ofclass-_____all
236. True bugs belongs to order .
237. Aphids are included in ord er of class_____Homoptera
238. Intelligent cultivation and use of fertilizers will aid in control of_____insects
239. True bugs are included in order of classHexapoda._____Hemipter
240. Grasshoppers damage_____both a and b
241. Electrical window screens are available for insects_____-electrocuting
242. There are 31 different orders of insects in the class_____Hexapoda
243. Drying of reduce the weevil injury._____grain
244. Which ofthe following class has onepair of legs perbodysegment?____Chilopoda
245. Coal-tar and pine are insect.killers_____repellent
246. Which of the following class has four pairs of jointed appendages?Arachnida
247.is the respiratory struc_____Air tubes
248. Insects have ---------- pair / pairs of wings._____Two
249. And the skeleton in arthropoda is made of----------which somewhatresembles horn in its nature_____Chitin
250. Insects are the members of---------- group of arthropods_____---Hexapoda
251.are the most successful , widespread and prolific members of the animal Kingdom._____Insects
252. Thecommon name of millipede is due tothe presence of large number of ---------- In the these animals._____Legs
253. The respiratory structure in Arachnids is called----------_____Book lungs
254. Which of the following is not the body part of insects ?_____Tail
255. The segments of the body of arachnids are grouped into---------- sections.__Two
256. Small diplopods are found boring into potatoes and sometimes wrongly called----------Wireworm
257. Which type of skeleton is found in arthropods?_____Exoskeleton
258. In Millipedes , eachsegment bears ---------- pair/pairs of legs._____Two
259. Millipedes are found on earth and breathe through ---------- --._____Book lung
260. Members of class ---------- are mainly water – inhibitinganimals._____Crustacea
261. Which of the following is NOT a distinctive feature of the arthropods?_____Radial symmetry
262. When the body segments are fused with the head they form----------_____Cephalothorax
263. Class Diplopoda include----------_____Millipede
264. Insects have ---------- pair / pairs of wings._____Two
265. which one of them is NOT the member of phylum Mollusca?_____Sea Urchin
266. Scorpions, spiders, mites, and ticks are the members of the group ---------- arthropods._____Arachnids
267. In the Hexapoda the segments of body are grouped in ---------- distinct sections._____Three
268. Insects have ---------- pair / pairs of antennae on their head._____Two
269. In insectsthere are ----------legs attached to the thorax._____6
270. The lobsters , cry-fish , scorpions , spiders, mites , ticks , centipedes , and millipedes are Members of ---------- group._____Insecta
271. Class diplopoda include----------_____Millipede
272. InMillipedes eachsegmentbears ---------- pair / pairs of legs._____Two
273. The common name of Millipede is due to the presence of large number of----------in These animals._____Legs

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274. Mammals belong to phylum ----------_____Chordata
275. Crustaceans breathe through ----------_____Gills and body surface
276. Which of the following organism has Book lung as respiratory organ?_____Scorpion
277. In ---------- the front leg on each side is modified to serve as a poison claw._____Chilopoda
278. When the body segments are fused with the head they form ----------_____Cephalothorax

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1. common name of millipede in due to reference of large no of legs in these animals.
2. The are areas of members b/t veins. Cell
3. The arthopode the body is composed of a series of more or less similar pieces or segment
4. Often some bugs body segmented are fused with the head to form a crustacean cephalopodan.

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5. In Archnicta generally have segment of the body grouped into tub section.
6. These structure and the joint line divided the cubicula into areas called plates or seclerites.
7. Sense organ are present in Antennae.
8. The arrangement and number of chief veins and their branch are of importance in identifying insect.
9. In colonial insects where of labor occurs division
10. In the female insect the eggs are produced in a pair of Llocated in the upper front part of the abdomen tubes
11. The group of insects that supply nourishment to the _before its birth are termed viviparous embry
12. The insects possesses a symmetrical body bconsisting of a series of segments. bilaterally
13. At the front end of the insect heart there is a tube without chambers or valves, called the aorta
14. In some insects who changed their habit to make holes for egg laying, the ovipositor being a sting
15. The group of insects that supply nourishment to the _before its birth are termed viviparous embryo
16. The arrangement and number of the chlef veins and their branches are of importance in identifying. Insects
17. Second ganglion in nervous lie behind or below the organ Spinal cor
18. The of the Hemimetabola appear not to have attracted sufficient attention to receive any special common names. nymph
19. But after digging into the earth a few inches, the insect forms a little chamber or cell in which to lie and generally lines this more or less densely with silk, probably to keep the earthen walls from falling in and crushing itearthen

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20. The young in the Hemimetabola thus limited are called those in the Paurometabola, nymphs naiads\ the young insect on escaping from the egg, though resembling its parent to some extent.hemimetabol
21. At one place on the surface is a minute opening or group of openings through the shell, called the micropyle for the purpose of together making holes in some object (the ground, wood, etc.) in which to deposit the eggs. Ovipositor
22. The nervous system of insects is located along the of the body quite near its under surface middle line
23. Its inner surface, forming the of the mouth, is often called the epipharynx Roof
24. The nervous system of insects is located along the of the body quite near its under surface middle line
25. in nature which must be removed like all wastes The cells by their activities produce not only carbon dioxide gas but also waste material from the body Nitrogenous all
26. But they also receive much of the gas produced by the activities of the cells and permit it to escape through the spiracles from the body carbon dioxid
27. **The ingrowth from the mouth is usually called the Fore intestine**
28. The wings are outgrowths from the body which vary much in size and form in different insects. Chitinous
29. its inner surface, forming the of the mouth, is often called the epipharynxRoof _mouth parts, being the more primitive and simple. Biting
30. These sutures and the joint lines divide the cuticula into areas called plates or Sclerites Sclerites
31. In the body of the adult insect there are apparently from segments 7 to 15
32. In the the segments of the body are grouped in three distinct sections. Insects groups are present in arthropoda. 4
33. The Common name of millipede is due to the presence of large number of in these animals.legs
34. The arthropod body is composed of different **Segments**
35. Main groups are present in Arthropods.**5**
36. The generally has the segmented body group into two sections.**Diplopoda**
37. The inner surface forming of the mouth is often called the epipharynx.**Roof**
38. Second ganglion in nerve lie or below the organ.**Esophagus**
39. The pathogens carried into insect body is transmission.**Physical**
40. Pesticides control is suitable only of **Pests**
41. Only_____entomologist imparts the secondary parasites.**Trained**
42. Average loss of crop due to insect damage is about %.**10**
43. Natural breeding method is the type of control.**Biological**
44. The Alimentary canal extended from the mounth through about the center of body to the anus at the hinder end.
45. The cell by their activities produce not only carbon dioxide gas but also waste material Nitrogenous in nature wich must be removed like all wastes from the body.
46. Below the or behind the esophagus is a second ganglion also in the head.
47. Those which lay eggs hatching later are termed oviparous.
48. Sense organ present in Antennae
49. Bug worm is example of **Lepidoptera**
50. In some flies the larvae shrinks within its larval skin and transforms, this skin is now called_

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Puparium

51. The introduction of parasite is **Artificial Control**

52. The destruction of weeds in of insects.**Reduction**

53. Cotton aphid belong to order **Homoptera**

54. Adult tripe causes silvery shine on the undersurface of **Leavel**

55. Brown plant hopper transmits disease in rice.**Virus**

56. Adult rice-ear are in color ear.**Yellow**

57. Egg of the shoot borer changes to larvae within days.**5-7**

58. Pinkworm damages the flower.**Rosette**

59. Shellac is a product.**Commercia**

60. Destruction of insectivore is fact.**Third**

61. IPM require planning.**Advanced**

62. The king of fiber **Cotton (Gossypium spp)**

63. Dusky cotton bug make the seed **Discolored and shrunken**

64. carry the insect to long distance.**Wipds**

65. The insects control is of type **2**

66. The timmimg or harvesting are important in **Practice**

67. Length of to-borer larvae mm.**30-35**

68. Thysanoptera has mouth **Rasping**

69. Potato leaf hopper plant sap.**Suck**

70. Mouth of neuropteran **Downward**

71. Coal tar is a **Repellent**

72. How many types of Artificial control of insects? **6**

are land animals breathing by air tubes opening on the sides of the body. Diplopoda

73. Often some of the bug; body segments are fused with the head to form a crustacean

Cephalothorax

74. The wings are outgrowths from the body. **Chitinous**

75. **_____**temperature is not effective for killing of insects. **Low**

76. **_____**attacks canes. **Grasshopper**

77. **_____**is a insects borne disease. **All**

78. Bollworm makes_____on boll. **Bore hole/hole**

79. Growing conditions for the insects is optimize control. **Cultural**

80. Introduction of a foreign pest is a **Cultural**

81. The_____extend from to body. **Elementary canal**

82. Those Organisms which lay egg hatching later is called **Oiparous**

83. In the_____the young insect on escaping from the egg, though resembling its parent.

Hemimetabola or Paurometabola

84. Electrical window screens are available for insects. **Electrocuting**

85. Adult rice ear are in color.**Yellow**

86. Natural Breeding method is a type of control.**Biological/Cultural**

87. Bug worm is an example of **Lepidoptera**

88. In some flies the larva shrinks within its larval skin and transforms, this skin, now called a

Puparium

89. Shellac is a product.**Commercial**

90. The_____are land animals breathing by air tubes opening on the sides of the body.**Diplopoda**

91. In the_____the segments of the body are grouped in three distinct section **Hexapoda**

92. In_____mouth parts, being the more primitive and simple.**Biting**

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93. The ingrowth from the mouth is usually called the **Fore intestine**

94. An enlarged portion of the vas deferens is often present, for the temporary storage of the sperms, and is termed

95. the **Seminal vesicle**

96. At one place on the surface is a minute opening or group of openings through the shell, called the **Micropyle**

97. The feeding periods between the molts (or ecdyses) are called **Instal's**

98. The arrangement and number of the chief veins and their branches are of importance in identifying **Insects**

99. The nymphs of the _ appear not to have attracted sufficient attention to receive any special common names. **Hemimetabola**

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